

Dilation Rigidity of Surreal Numbers and Omnific Integers

Autonomous Mahler equations, finite support,
sharp algebraic degrees, and a higher-order hierarchy

Research article prepared with ChatGPT
for Vladimir Reshetnikov

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Built from one manuscript of batch 30 of the collection, pinned to repository commit [b895e86](#)

Abstract

For an integer $d \geq 2$, exponent dilation sends $\sum a_\gamma t^\gamma$ to $\sum a_\gamma t^{d\gamma}$. We prove that a nonconstant Hahn series algebraically dependent with its own dilate over the constant field must belong to a single ordinary Laurent series subfield. This holds in characteristic zero over every ordered abelian value group, without an Archimedean or divisibility assumption. If $F(y, S_d y) = 0$, the necessary ramification is at most $\deg_y F$. The proof combines Newton–Puiseux branches with a formal Böttcher coordinate and a least-support-term argument; every evaluation is justified by strong Hahn summability, not by a contraction principle.

For the negative-support ring this becomes an exact classification: first-order autonomous algebraic dependence occurs precisely for polynomials in one monomial. The least degree in the dilated variable is exactly the degree in the primitive monomial. Consequently every infinite-support omnific integer is algebraically independent of each of its nontrivial integer dilates. We also obtain a finite-profile description for a fixed algebraic equation, an exact Newton-weight existence test, complete rational-map solutions, and a sharp infinitesimal-tail obstruction to omnific solutions. Higher order behaves differently: an infinite-support omnific integer has exact autonomous order two, and examples realize every finite order. A research agenda identifies the remaining classification, effectivity, positive-characteristic, and formalization problems.

Research status. The proposed contributions are the arbitrary-rank classifications and their consequences, not the classical Newton–Puiseux or Böttcher theorems. Complete written arguments are supplied relative to the stated classical inputs. The results have not been refereed or Lean-verified, and priority is not certified. No named published conjecture is claimed solved. Exact finite checks accompany the article but do not verify the infinite theorems. An AI-assisted research draft. Every label carries the prefix `adr:`. Conventions, the relation to the collection as it now stands, the ledger of non-claims and the provenance are in [sections 1.4, 1.5 and 18](#) and [Appendix C](#); text added when the manuscript joined the collection is marked `[write]` where it sits inside the source’s text.

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1 Research position and the questions answered

An omnific integer can have a transfinite normal form, infinitely many nonzero coefficients, and exponents at mutually incomparable Archimedean scales. A polynomial in one monomial has none of this support complexity. The principal question of this article is whether an algebraic relation between a number and its own exponent-dilated copy can distinguish these situations.

The dilation is

$$S_d \left(\sum_{\alpha} a_{\alpha} \omega^{g_{\alpha}} \right) = \sum_{\alpha} a_{\alpha} \omega^{d g_{\alpha}}. \quad (1.1)$$

It is a field automorphism on **No** and **No**[i], and fixes ordinary constants. It is *not* the map $y \mapsto y^d$. For example,

$$S_2(\omega + 1) = \omega^2 + 1, \quad (\omega + 1)^2 = \omega^2 + 2\omega + 1.$$

The question is about polynomial equations over the ordinary constant field, with S_d additionally available. It is not a statement that an infinite omnific integer is algebraic over $\mathbb{R}$.

1.1 The main answer in omnific notation

For $x \in \mathbf{Oz} \setminus \mathbb{Z}$ and $d \geq 2$, we prove the equivalence

$$\begin{aligned} & x \text{ and } S_d x \text{ are algebraically dependent over } \mathbb{R} \\ \iff & x = P(\omega^{\delta}) \text{ for some } \delta > 0, \quad P \in \mathbb{R}[T], \quad P(0) \in \mathbb{Z}. \end{aligned} \quad (1.2)$$

The same statement holds for Gaussian omnific integers with $\mathbb{C}$ and $\mathbb{Z}[i]$ in place of $\mathbb{R}$ and $\mathbb{Z}$. Thus an infinite normal-form support is an unconditional obstruction to *every* first-order autonomous algebraic relation. A finite support can also obstruct a relation: its positive exponents must be commensurable, not merely comparable in size.

There is an exact quantitative refinement. Normalize δ so that the nonzero positive exponents of x generate $\delta\mathbb{Z}$. Then, writing $x = P(\omega^{\delta})$ and $M = \deg P$,

$$\min\{\deg_Y F : 0 \neq F \in \mathbb{R}[X, Y], F(x, S_d x) = 0\} = M. \quad (1.3)$$

In particular, the answer is independent of the chosen integer $d \geq 2$. This is a classification of a substantial class of equations, rather than one specially selected identity.

1.2 Relation to the repository

The repository was inspected at commit

b895e8672990e8b5a56f97dd7246f9dd5f86c771.

The main points of contact are the report on a single dilation [2], the dynamics report [3], and the omnific and Hahn-support infrastructure catalogued in the repository [1]. In particular, the single-dilation report already uses the recognizer $S_2 w = w^2$ for canonical monomials and treats linear and multiplicative difference equations. That recognizer is a premise to extend, not a novelty to reclaim. We reprove its elementary characteristic-zero version for general d .

The present extension is from this one special equation to *every* constant-coefficient polynomial relation $F(y, S_d y) = 0$, including ramified algebraic correspondences. It supplies support bounds, an exact omnific

classification, and sharp degrees. The inspected repository excerpts do not supply this classification. That comparison is not an assertion that every line of every repository manuscript has been independently reviewed. [write] This subsection is the source's, written at its pin. The relation to the collection as it now stands, with exact labels, is in section 1.5.

1.3 Literature and the novelty boundary

Hahn summation and its support calculus are classical [4]. Newton–Puiseux factorization is classical; Mannaa–Coquand give a modern constructive treatment [5]. Formal Böttcher coordinates are also classical; the needed characteristic-zero theorem is recorded explicitly by Salerno–Silverman [6]. We use these inputs openly and give an explicit coordinate construction adapted to strong Hahn evaluation.

There is an established literature on autonomous Mahler equations, including Nishioka–Nishioka [7], and on nonlinear formal Mahler solutions [9]. Chyzak–Dreyfus–Dumas–Mezzarobba [10] and Faverjon–Roques [11] address linear Mahler equations and generalized series. The later works of Faverjon–Poulet [12] and Faverjon–Roques [13] develop basis algorithms and generalized Mahler-series structure for linear systems; they are cited here for context from their abstracts, not used as proof dependencies. These works prevent any claim that Mahler equations, Böttcher coordinates, or non-Puiseux solutions are new topics. The contribution proposed here is the *arbitrary-rank support-collapse theorem*, the resulting exact omnific criterion, and their quantitative synthesis. [write] The full text of [7] was read when the manuscript joined the collection. It records a 1983 criterion of Mahler [8] for the existence of a convergent power-series solution of a constant-coefficient relation $P(f(z), f(z^d)) = 0$, a rank-one, positive-support antecedent of the existence statements of section 7; see Appendix C.3.

For omnific arithmetic, L’Innocente–Mantova [14] provide the relevant published factorization framework. Our result is not a factorization theorem and does not settle the remaining general factorization conjectures. The user’s encyclopedia link [16] is useful orientation, but none of the proofs relies on an encyclopedia assertion in place of the classical inputs above.

1.4 Notation and conventions in this report

[write] This subsection and the next were written when the manuscript joined the collection. No symbol of the source was renamed and no normalization was changed; the meanings below are fixed once for the whole report, and paths such as `surcomplex/single-dilation-hahn-support` are relative to `docs/`.

- *The dilation.* S_d is the exponent dilation (2.1), with $d \geq 2$ an ordinary integer; in section 11 and section 14 also S_q for rational $q > 1$. It is the map that the single-dilation report writes S_q at $q = d$ (`dsup:sec:conventions`) and that the surcomplex-automorphisms report writes S_a at $a = d$ (`saut:eq:dilation`). On **No** it is the canonical Hahn lift of the value-group dilation $\gamma \mapsto d\gamma$ in the exponential-automorphism report (`prop:hahn-lift`; that report writes the dilation T_r and its lift $\widehat{T}_r$). The tempting false readings, beside the true one: S_d is not $y \mapsto y^d$ (the two agree exactly on the monomials of lemma 3.3); it is not the argument dilation $f(z) \mapsto f(\lambda z)$ of formal functions in the holonomic-rigidity report (`hol:sec:qneccessity`); and it is not an exponential automorphism (section 1.5).
- *Fields and rings.* $K_\Gamma(k) = k((t^\Gamma))$ is a set-sized full Hahn field over an arbitrary field k , of characteristic zero wherever the main results are stated; the notation guide `NOTATION.md` writes K_Γ for the case $k = \mathbb{C}$ and F_Γ for $k = \mathbb{R}$. The ring $A_\Gamma(k)$ of (2.3) is the *negative-support* ring, support in $\Gamma_{\leq 0}$; it is not the valuation ring $\mathcal{O}$. Under (2.4) it is the ring of forms with nonnegative growth support, which the notation guide records as the Diophantine report’s $\mathcal{A}_k(\Gamma)$. The symbol $A_\Gamma(k)$ is local to this report. **Oz** and **Oz**[i] are as in (2.5), in agreement with the guide’s **Oz** = $\mathbb{Z} \oplus \Pi$.

- *Order.* The autonomous algebraic order of section 13 is the least r for which $y, S_d y, \dots, S_d^r y$ are algebraically dependent over k . It is not the order of a linear Mahler equation, and not the differential order of the holonomic-rigidity report. ord_Z is the Z -adic order of an ordinary Laurent series.
- *Local letters.* $M(y)$ is the primitive degree of section 9; it is not the monomial formula $M(m)$ of the single-dilation report (`dsup:eq:monomialformula`). The letters M (an auxiliary series in the proof of lemma 5.1) and M_c (a chart in theorem 11.1) are local. M is also a polynomial degree in theorems 15.1 and 15.2. D is an integer degree in lemma 4.1 and theorems 6.1, 9.2, 15.1 and 15.2 and an auxiliary series in the proof of lemma 5.1; it never denotes a derivation. H_a is the normalized inverse Böttcher coordinate of proposition 5.2, an ordinary power series; it is not the convex subgroup H_a of the notation guide. L is a series in lemma 5.1 and in Appendix B, and a field in corollary 7.2 and theorem 9.2. T is a monomial variable, $t^{-\delta}$ or ω^δ , or an ordinary polynomial variable, and $T_j = \omega^{g_j}$ in theorem 13.3.
- *Strong* means preserving Hahn summation, not convergence of partial sums (section 3). *Cyclically supported* means support in one cyclic subgroup $\delta\mathbb{Z}$, bounded below (section 2).

1.5 Relation to other reports in the collection

[write] The source compared itself with selected reports at its pin (Appendix C.2). Checked against the collection at the placement commit, the relations are these.

Single dilation (`surcomplex/single-dilation-hahn-support`, prefix `dsup:`). This is the nearest report: the same operator and the same recognizer.

1. *Recognizer.* Lemma 3.3 at $d = 2$ is `dsup:lem:monomialrecognition`, where $a = a^2$ forces coefficient one, with the same least-term proof; the general- d statement is the source's. The equality $\text{Fix}(S_d) = k$ of lemma 3.2 is `dsup:eq:dilationfixed`. The report there also has the power-equation recognizer `dsup:lem:rationalrecognition` (item 3 below).
2. *Linear and multiplicative first-order equations* are treated there (`dsup:thm:dilationcohom`). The boundary equation $S_d u_d - u_d = \omega^d$ of section 14 is, at $d = 2$, the Conway form of that report's example `dsup:ex:opposite`: under (2.4) its second series becomes $y = \sum_{n \geq 1} \omega^{2^{-n}}$ with $S_2 y - y = \omega$, and $u_2 = S_2 y$, so applying S_2 gives $S_2 u_2 - u_2 = \omega^2$. For general d the same telescoping applies. That report presents the example as an illustrative calculation; the source credits the series pattern to [10, 9]. Only the exact autonomous order of theorem 13.2 is claimed here.
3. *Nonintegral rational dilations.* The source leaves implicit relations $F(y, S_q y) = 0$ with $q \in \mathbb{Q}_{>1} \setminus \mathbb{Z}$ open (section 14 and the question on nonintegral rational dilation in section 16). The single-dilation report already exhibits nonconstant solutions of such relations. For divisible Γ , $q = a/b$ with positive integers a, b and $n = |a - b|$, `dsup:lem:rationalrecognition` shows that the nonzero solutions of $(S_q x)^b = x^a$, that is of $F(X, Y) = Y^b - X^a$, are exactly the elements of $\mu_n(k) t^\Gamma$. These solutions are monomials, so they satisfy the cyclic-support conclusion trivially. There is no contradiction with theorem 11.1: since $q \notin \mathbb{Z}$ one has $b \geq 2$, and t^{qg} is not a rational function of t^g for $g \neq 0$. This is a partial datum for the question, not an answer: whether every solution of every such relation has controlled support remains open, and the caution at the end of section 14 stands.
4. *Decidability.* The undecidability of that report (`dsup:thm:undecidable`) concerns the full first-order theory of the expanded field; theorem 15.2 decides one specified equation. The source says there is no contradiction (section 14), and there is none.

5. *Its questions and non-claims.* None of the three questions of `dsup:sec:questions` is answered. In particular its question on definability from other exponent automorphisms asks which ordered exponent automorphisms θ admit polynomial equations recognizing canonical monomials; this report treats only $\theta =$ multiplication by an integer $d \geq 2$. Its non-claim that no solvability theorem for arbitrary variable-coefficient Mahler systems is inferred (item 7 of `dsup:sec:nonclaims`) stands: this report is about constant-coefficient nonlinear relations, and section 14 shows that the collapse fails with nonconstant coefficients.

Surcomplex field automorphisms (`surcomplex/surcomplex-field-automorphisms`, prefix `saut:`). S_d is its S_a at $a = d$ (`saut:eq:dilation`), built by its monomial construction (`saut:thm:monomial`). None of the questions of `saut:sec:questions` concerns algebraic relations between an element and its dilate, and none is addressed.

Exponential automorphism rigidity (`surreal/exponential-automorphism-rigidity`, bare labels). On **No**, S_d is the canonical lift `prop:hahn-lift` of the value-group dilation by d , and `thm:dilation` shows that for rational $r \neq 1$ the value-group dilation by r has no exponential lift at all. So S_d does not commute with the surreal exponential. That report's subject, which value-group automorphisms lift to exponential automorphisms (`q:image`), is not this report's subject: nothing here bears on it, and the source's last question, on the interaction of the finite-profile classes with the surreal exponential, remains open and is consistent with that theorem.

Holonomic rigidity (`surcomplex/holonomic-rigidity-for-entire-hahn-functions`, prefix `hol:`). Its dilations are argument dilations $f(z) \mapsto f(qz)$ of entire Hahn functions by a scalar q of the field, in linear q -difference equations (`hol:main:q`, `hol:sec:qnecessity`). They are not the exponent dilations of this report, whose equations are between numbers, not functions. No statement of that report is used or extended, as the single-dilation report also records.

Dynamics (`surcomplex/dynamics-and-normal-forms`). The source lists it as a point of contact. It studies multipliers, linearization and normal forms of germs; the superattracting germs and the Böttcher coordinate of section 5 occur in no other report of the collection, and no statement of that report is used.

Formalization. No `adr:` label has a Lean mapping in `FORMALIZATION.md`; no theorem of this report is Lean-verified.

2 Hahn fields, normal forms, and the two support directions

Definition 2.1. Let k be a field and Γ an ordered abelian group. The full Hahn field $K_\Gamma(k) = k((t^\Gamma))$ consists of formal sums

$$y = \sum_{\gamma \in \Gamma} a_\gamma t^\gamma, \quad a_\gamma \in k,$$

whose support is a well-ordered subset of Γ . For $y \neq 0$, set $v(y) = \min \text{supp } y$ and let $\text{lc}(y)$ be the coefficient there. Put $v(0) = +\infty$.

Throughout the main classification, $\text{char } k = 0$ and d is an ordinary integer at least two. No order on k is required. Dilation

$$S_d\left(\sum a_\gamma t^\gamma\right) = \sum a_\gamma t^{d\gamma} \quad (2.1)$$

is a strong injective field endomorphism; it is an automorphism when Γ is d -divisible. Its elementary valuation law is

$$v(S_d y) = d v(y). \quad (2.2)$$

The valuation ring and its maximal ideal are

$$\mathcal{O} = \{y : v(y) \geq 0\}, \quad \mathfrak{m} = \{y : v(y) > 0\}.$$

For $y \in \mathcal{O}$, its residue is its coefficient at zero, an element of k . In contrast, define

$$A_\Gamma(k) = \{y \in K_\Gamma(k) : \text{supp } y \subseteq \Gamma_{\leq 0}\}. \quad (2.3)$$

This is a subring, but it is *not* the valuation ring. A nonconstant element of $A_\Gamma(k)$ has negative valuation. The absence of *positive* exponents, not just a bound on the leading exponent, will be decisive.

Definition 2.2. For $\delta \in \Gamma_{>0}$, the notation $k((t^\delta))$ means the image of the ordinary Laurent series field $k((Z))$ under $Z \mapsto t^\delta$. Thus its elements have support contained in $\delta\mathbb{Z}$, bounded below in the integer indexing. A series with this property is called *cyclically supported* here.

Cyclic support is stronger than support in one Archimedean class and stronger than a one-dimensional rational span. The set $\{-1, -1/2, -1/4, \dots\}$ spans a one-dimensional $\mathbb{Q}$ -space but is not contained in any cyclic subgroup of $\mathbb{Q}$.

2.1 Translation to surreal numbers

Conway normal forms use decreasing powers of ω . The correspondence with the increasing Hahn convention is

$$t^\gamma \longmapsto \omega^{-\gamma}. \quad (2.4)$$

For any set-sized ordered additive subgroup $\Gamma \subseteq \mathbf{No}$, it identifies $\mathbb{R}((t^\Gamma))$ with the corresponding normal-form subfield of $\mathbf{No}$; complexifying gives $\mathbb{C}((t^\Gamma)) \subseteq \mathbf{No}[i]$. Every individual surreal or surcomplex number is contained in such a workspace: take the divisible subgroup generated by its set-sized support. The usual normal-form arithmetic is the algebraic input here [14, 1].

The omnific integers have the description

$$\mathbf{Oz} = \left\{ n + \sum_{\alpha < \lambda} r_\alpha \omega^{g_\alpha} : n \in \mathbb{Z}, r_\alpha \in \mathbb{R}, g_\alpha > 0, (g_\alpha)_{\alpha < \lambda} \text{ strictly decreasing} \right\}, \quad (2.5)$$

with set-sized support and zero coefficients omitted. Thus the only additional restriction beyond the real negative-support ring is the integrality of the coefficient at zero. For $\mathbf{Oz}[i]$ the coefficients are complex and that constant lies in $\mathbb{Z}[i]$.

2.2 Size and topology conventions

All field-theoretic proofs below take place in a set-sized Hahn field. The statements for **No** and **No**[i] are then elementwise consequences. No proper-class polynomial ring, class-sized tensor product, or class-indexed analytic sum is required.

Also, $v(x) > 0$ need not imply that $v(x^n)$ tends cofinally to infinity in Γ . For example, in the lexicographically ordered group $\mathbb{Q} \oplus \mathbb{Q}$, the multiples of $(0, 1)$ never reach $(1, 0)$. Nevertheless the geometric series in x is strongly summable. This separation between support summability and topological convergence is essential throughout.

3 The support calculus and a monomial recognizer

A family (f_i) of Hahn series is *strongly summable* when the union of its supports is well ordered and every exponent occurs in the support of only finitely many family members. Its sum is then defined coefficientwise.

Lemma 3.1 (Formal evaluation). *Let $x \in \mathfrak{m}$. Every formal series $f(Z) \in k[[Z]]$ can be evaluated at x as a strong Hahn sum. This gives a k -algebra homomorphism. For $x \neq 0$ it extends to an injective map $k((Z)) \rightarrow K_\Gamma(k)$, and*

$$v(f(x)) = \text{ord}_Z(f) v(x) \quad (f \neq 0). \quad (3.1)$$

Formal composition is respected whenever the inner series has zero constant coefficient. Moreover,

$$S_d(f(x)) = f(S_d x). \quad (3.2)$$

Proof. The Neumann support lemma says that the monoid generated by a well-ordered subset of $\Gamma_{>0}$ is well ordered and that each exponent has only finitely many representations by finite words from that subset, allowing all word lengths [4]. Apply it to $\text{supp } x$. It justifies the family of powers and the regroupings for products and formal composition. There are only finitely many negative powers in a Laurent series, so these cause no extra infinite-sum issue.

If the initial term of f is aZ^m , its evaluation has initial valuation $m v(x)$: every later power has strictly larger valuation. This proves (3.1) and injectivity. Exponent dilation preserves well ordering, finite occurrence, and coefficientwise sums, proving (3.2). $\square$

Lemma 3.2 (Constants are relatively algebraically closed). *Every element of $K_\Gamma(k)$ algebraic over k belongs to k . In particular, every nonconstant Hahn series is transcendental over k , and $\text{Fix}(S_d) = k$.*

Proof. A nonzero algebraic element z satisfies a polynomial with nonzero constant and leading coefficients in k . If $v(z) > 0$, the constant term is the unique term of least valuation; if $v(z) < 0$, the highest-degree term is. Both are impossible. Hence $v(z) = 0$. Let $c \in k$ be its residue. If $z - c \neq 0$, this is an algebraic element of positive valuation, again impossible.

For a nonconstant fixed element y , use $x = y - c$ if $v(y) \geq 0$, with c its residue, and $x = 1/y$ if $v(y) < 0$. In both cases $x \neq 0$ has positive valuation and is fixed. The equation $d v(x) = v(x)$ contradicts $d > 1$ in an ordered group. $\square$

Lemma 3.3 (Monomial recognizer). *Suppose $w \neq 0$ and $S_d w = w^d$. Then*

$$w = at^\eta, \quad a \in k^\times, \quad a^{d-1} = 1, \quad \eta \in \Gamma. \quad (3.3)$$

Conversely every such monomial is a solution.

Proof. Write $w = at^\eta(1 + \varepsilon)$ with $\varepsilon \in \mathfrak{m}$. Initial coefficients give $a = a^d$. Dividing the original identity by $at^{d\eta}$ gives

$$1 + S_d \varepsilon = (1 + \varepsilon)^d.$$

If $\varepsilon \neq 0$ and $\rho = v(\varepsilon) > 0$, then the nonconstant part of the left side has valuation $d\rho$, whereas that of the right side has valuation ρ : its initial coefficient is multiplied by the nonzero scalar d . Since $d\rho > \rho$, this is impossible. $\square$

The least-term proof eliminates possible tails at *all* other scales at once. It neither examines the terms sequentially nor assumes that an iteration approaches its limit in the ambient valuation topology. The $d = 2$ instance is the repository's monomial recognizer [2]. [write] That is `dsup:lem:monomialrecognition`, proved there by the same least-term comparison; the equality $\text{Fix}(S_d) = k$ of lemma 3.2 is `dsup:eq:dilationfixed` (section 1.5).

4 The classical ramification input and its degree bound

We use the Newton–Puiseux theorem in the following standard form: if k is algebraically closed of characteristic zero, then

$$\bigcup_{e \geq 1} k((U^{1/e}))$$

is an algebraic closure of $k((U))$ [5]. Thus every nonzero polynomial in $k((U))[V]$ splits into linear factors there. The next lemma spells out both the quantitative point and its transport into a Hahn field.

Lemma 4.1 (Evaluated Puiseux branch). *Assume k algebraically closed, Γ divisible, and $x \in K_\Gamma(k)$ with $v(x) = \gamma > 0$. Let $0 \neq G(U, V) \in k[U, V]$ have V -degree D , and suppose $G(x, z) = 0$. Then there exist an integer $1 \leq e \leq D$, an e th root u of x in $K_\Gamma(k)$, and $\phi(Z) \in k((Z))$ such that*

$$z = \phi(u), \quad G(Z^e, \phi(Z)) = 0. \quad (4.1)$$

Moreover $v(u) = \gamma/e$, and $v(z) = \text{ord}_Z(\phi) \gamma/e$ when $z \neq 0$.

Proof. Apply Newton–Puiseux to $G(U, V)$ over $k((U))$. In a common extension $k((U^{1/N}))$ it factors as a nonzero leading coefficient times finitely many factors $V - \psi_j(U^{1/N})$.

An N th root of x exists in the Hahn field without using its full algebraic closedness: write $x = at^\gamma(1 + \varepsilon)$, choose $b^N = a$, and use

$$bt^{\gamma/N}(1 + \varepsilon)^{1/N},$$

where the binomial series is evaluated by lemma 3.1. Evaluation at this root sends the Puiseux factorization to a genuine factorization in $K_\Gamma(k)$. Its leading coefficient is nonzero by the injectivity in lemma 3.1. Consequently z equals one of the evaluated roots.

Choose the smallest denominator e for that root. Its series can be written as $\phi(U^{1/e})$, with $e \mid N$; the corresponding power of the chosen N th root gives u . Minimality means that the automorphisms $U^{1/e} \mapsto \zeta U^{1/e}$, for $\zeta^e = 1$, give exactly e distinct conjugates of this series. Indeed a stabilizing ζ must satisfy $\zeta^j = 1$ for every exponent j occurring in ϕ , and minimality says their common divisor with e is one. All these conjugates are roots of G . Hence $e \leq D$. The valuation statements follow from lemma 3.1. $\square$

Remark 4.2. The bound is on the denominator of *one selected branch*, not on a common denominator for every root. A common splitting denominator can be larger than D . Confusing these two denominators would invalidate the sharp degree theorem below.

5 A support-safe Böttcher construction

The formal-coordinate theorem in this section is classical [6, Proposition 1]. We include a construction so that no unstated convergence hypothesis enters its Hahn-field application. All sums in the construction are first formed in the ordinary formal ring $k[[Z]]$.

Lemma 5.1 (Normalized formal coordinate). *Let $h(Z) = Z^d U(Z)$, where $U \in 1 + Zk[[Z]]$ and $\text{char } k = 0$. There is a unique $b(Z) \in Z + Z^2 k[[Z]]$ with*

$$b(h(Z)) = b(Z)^d. \quad (5.1)$$

Its compositional inverse H satisfies

$$H(Z^d) = h(H(Z)). \quad (5.2)$$

Proof. Put

$$L(Z) = \sum_{n \geq 0} \frac{1}{d^{n+1}} \log U(h^{on}(Z)), \quad b(Z) = Z \exp L(Z). \quad (5.3)$$

The n th summand has order at least d^n unless it is zero, so each coefficient receives only finitely many contributions. Reindexing gives

$$dL(Z) = \log U(Z) + L(h(Z)).$$

Therefore

$$b(h(Z)) = Z^d U(Z) \exp L(h(Z)) = Z^d \exp(dL(Z)) = b(Z)^d.$$

The linear coefficient of b is one, so it has a unique compositional inverse, which gives (5.2).

For uniqueness, any normalized coordinate has the form $Z \exp M(Z)$. The difference $D = L - M$ satisfies $D(h) = dD$. A nonzero D of positive order would have different orders on these two sides, namely $d \text{ ord } D$ and $\text{ord } D$. Thus $D = 0$. $\square$

Proposition 5.2 (All Hahn solutions of a superattracting germ). *Let $h(Z) = AZ^d + O(Z^{d+1}) \in k[[Z]]$, $A \neq 0$, and let $x \neq 0$ have positive valuation. Then $S_d x = h(x)$ is solvable with prescribed leading coefficient $a \in k^\times$ and valuation $\delta \in \Gamma_{>0}$ if and only if*

$$Aa^{d-1} = 1. \quad (5.4)$$

For each such pair there is exactly one solution:

$$x = aH_a(t^\delta), \quad (5.5)$$

where H_a is the normalized inverse Böttcher coordinate of $h_a(Z) = a^{-1}h(aZ)$.

Proof. Leading coefficients force (5.4). With this condition, $h_a(Z) = Z^d + O(Z^{d+1})$. Write $u = x/a$, and let b_a be the coordinate of lemma 5.1. Strong evaluation and its commutation with S_d give

$$S_d(b_a(u)) = b_a(S_d u) = b_a(h_a(u)) = b_a(u)^d.$$

The left-hand argument $b_a(u)$ has valuation δ and leading coefficient one. By lemma 3.3 it is exactly t^δ . Inversion yields (5.5) and uniqueness. Conversely, (5.2), evaluated strongly at t^δ , verifies that this expression is a solution. $\square$

This is an exhaustive statement in the full Hahn field. The coordinate series is ordinary, but the element to which it is applied may initially have arbitrary support. The recognizer proves that every extra support scale disappears after the coordinate change.

6 The arbitrary-rank support-collapse theorem

Theorem 6.1 (First-order autonomous support rigidity). *Let k be any field of characteristic zero, Γ any ordered abelian group, and $d \geq 2$. Suppose $y \in K_\Gamma(k) \setminus k$ and*

$$F(y, S_dy) = 0, \quad 0 \neq F \in k[X, Y]. \quad (6.1)$$

Define a positive-valuation coordinate

$$x = \begin{cases} y - c, & v(y) \geq 0, \quad c = \text{res}(y), \\ 1/y, & v(y) < 0, \end{cases} \quad \gamma = v(x) > 0. \quad (6.2)$$

Then for some integer e with $1 \leq e \leq \deg_Y F$, one has

$$\delta = \gamma/e \in \Gamma, \quad y \in k((t^\delta)). \quad (6.3)$$

In particular, the support is finite or has order type ω and is contained in a cyclic subgroup of Γ .

Proof. First assume k algebraically closed and Γ divisible. Let $D = \deg_Y F$. Since y is transcendental over k , $D \geq 1$. The equation becomes $G(x, S_dx) = 0$ for a nonzero polynomial G with $\deg_V G \leq D$. In the finite case use

$$G(U, V) = F(c + U, c + V).$$

In the infinite case, if $a = \deg_X F$, use

$$G(U, V) = U^a V^D F(U^{-1}, V^{-1});$$

monomial factors can be removed. These substitutions give nonzero polynomials, and x, S_dx are nonzero.

Apply lemma 4.1. There are $1 \leq e \leq D$, an element z with $z^e = x$, and a Laurent series ϕ such that

$$S_dx = \phi(z), \quad v(z) = \gamma/e. \quad (6.4)$$

But $v(S_dx) = d\gamma$. Consequently

$$\text{ord}_Z \phi = de, \quad \phi(Z) = BZ^{de}(1 + u(Z)), \quad B \neq 0, \quad u \in Zk[[Z]].$$

Choose an e th root of B and form a formal e th root of this series. Every such root has the form

$$h(Z) = AZ^d(1 + u(Z))^{1/e} \in k[[Z]], \quad A^e = B.$$

Since $(S_dz)^e = S_dx = \phi(z)$, the ratio between S_dz and one evaluated root $h(z)$ is an e th root of unity. Multiplying A by that root of unity gives an h with the exact identity

$$S_dz = h(z). \quad (6.5)$$

By proposition 5.2, $z = aH(t^{\gamma/e})$ for an ordinary formal power series H and a nonzero constant a . Thus $x = z^e$ belongs to $k((t^{\gamma/e}))$, as does $y = c + x$ or $y = 1/x$. This proves the assertion in the auxiliary algebraically closed, divisible setting.

For general k and Γ , embed the Hahn field coefficientwise and exponentwise into

$$\bar{k}((t^{\Gamma_{\mathbb{Q}}}), \quad \Gamma_{\mathbb{Q}} = \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$$

with its natural ordered divisible hull. The proof gives an integer $e \leq D$ and support contained in $(\gamma/e)\mathbb{Z}$. The intersection

$$\Gamma \cap (\gamma/e)\mathbb{Z} = m(\gamma/e)\mathbb{Z}$$

for a positive integer m dividing e : it is a nonzero subgroup of a cyclic group and contains γ . Put $e' = e/m$. Then $1 \leq e' \leq D$, $\gamma/e' \in \Gamma$, and the original support is contained in $(\gamma/e')\mathbb{Z}$. Its coefficients are still in k , by uniqueness of Hahn coefficients. The support is bounded below in that cyclic group. Renaming e' as e proves (6.3). $\square$

Remark 6.2 (What the theorem does not assert). Membership in a cyclic Laurent subfield is a necessary condition for (6.1), not a claim that every ordinary Laurent series satisfies some autonomous Mahler equation. Nor does the support conclusion turn strong evaluation into ambient valuation convergence. The exponent group can have arbitrarily high rank even when the individual solution is forced into one cyclic subgroup.

7 Finite profiles, coefficient descent, and an existence test

The proof gives more than a necessary support condition. For a fixed polynomial equation it supplies a finite collection of ordinary formal shapes; the only remaining continuous scale parameter is the monomial at which a shape is evaluated.

Theorem 7.1 (Finite-profile classification). *Let k be algebraically closed of characteristic zero, $d \geq 2$, and $0 \neq F \in k[X, Y]$. There is a finite, possibly empty collection*

$$\mathcal{P}(F, d) \subset k((Z)) \setminus k$$

such that for every nonzero divisible ordered group Γ ,

$$\{y \in K_\Gamma(k) \setminus k : F(y, S_d y) = 0\} = \{f(t^\delta) : f \in \mathcal{P}(F, d), \delta \in \Gamma_{>0}\}. \quad (7.1)$$

Each profile is constructed from a Puiseux branch and a normalized Böttcher coordinate. A profile at infinity has pole order at most $\deg_Y F$. The list need not be free of duplicates.

Proof. Remove all factors $Y - X$ from F , obtaining F_0 . A nonconstant solution cannot satisfy $S_d y = y$ by lemma 3.2, so this removal does not change the nonconstant solution set. The polynomial $F_0(X, X)$ is not identically zero. Consequently there are only finitely many possible finite residues c , namely its roots.

For each such c , form $G(U, V) = F_0(c + U, c + V)$. Also form the reciprocal polynomial at infinity as in the proof of theorem 6.1. For each of these finitely many polynomials, choose its finitely many Puiseux roots $V = \phi(U^{1/e})$. Keep only the nonzero branches satisfying $\text{ord}_Z \phi = de$. For each kept branch, there are finitely many formal e th roots

$$h(Z)^e = \phi(Z), \quad h(Z) = AZ^d + O(Z^{d+1}),$$

and finitely many constants a with $Aa^{d-1} = 1$. Let H_a be the coordinate in proposition 5.2. Put on the list

$$f(Z) = c + (aH_a(Z))^e \quad \text{or} \quad f(Z) = (aH_a(Z))^{-e}, \quad (7.2)$$

according to the finite or infinite chart.

The proof of theorem 6.1 shows that every nonconstant solution has one of these forms after evaluation at a positive monomial. Conversely let $z = aH_a(t^\delta)$. Then $S_d z = h(z)$, so with $x = z^e$,

$$S_d x = h(z)^e = \phi(z), \quad G(x, S_d x) = 0.$$

Undoing the chart gives the original equation. The pole-order bound is the branch denominator bound $e \leq \deg_Y F$. $\square$

Corollary 7.2 (Coefficient descent). *Let $k_0 \subseteq L$ be characteristic-zero fields and $0 \neq F \in k_0[X, Y]$. If $y \in L((t^\Gamma)) \setminus L$ satisfies $F(y, S_dy) = 0$, then every Hahn coefficient of y is algebraic over k_0 . In particular, an equation over $\overline{\mathbb{Q}}$ admits no nonconstant solution having a coefficient transcendental over $\overline{\mathbb{Q}}$.*

Proof. Work in an algebraic closure $\overline{L}$ and let k' be the algebraic closure of k_0 inside $\overline{L}$. Construct the finite profiles over k' . Their Puiseux factorizations still split the relevant polynomials over $\overline{L}((U))$; the roots of unity and leading-coefficient roots are already in k' . Thus the same exhaustion proof gives profiles with all coefficients in k' , even for a solution originally in the larger coefficient field. Evaluation at a monomial changes exponents, not coefficients. The original coefficients lie in $L \cap k'$. $\square$

This is a descent statement for a *fixed coefficient field of the equation*. It does not contradict the existence of $a\omega^\delta$ as a first-order solution over $\mathbb{R}$ for arbitrary real a : the corresponding equation generally contains a among its coefficients.

Theorem 7.3 (Exact local Newton-weight criterion). *Let k be algebraically closed of characteristic zero and $\Gamma \neq 0$ divisible. For*

$$0 \neq G(U, V) = \sum g_{ij} U^i V^j \in k[U, V],$$

there is a nonzero $x \in \mathfrak{m}$ satisfying $G(x, S_dx) = 0$ if and only if

$$\min\{i + dj : g_{ij} \neq 0\} \quad \text{is attained by at least two monomials.} \quad (7.3)$$

Proof. If $v(x) = \gamma > 0$, the valuation of a term is $(i + dj)\gamma$. A unique least term cannot cancel, proving necessity.

For sufficiency use the ordinary Newton polygon, or equivalently the Puiseux factorization. The weighted initial polynomial

$$I(W) = \sum_{i+dj=w} g_{ij} W^j, \quad w = \min(i + dj),$$

has at least two terms. Distinct active monomials have distinct j . After its monomial factor is removed, it has a nonzero constant term and positive degree, hence a nonzero root over k . Factoring G into Puiseux linear factors shows that these nonzero roots are exactly the leading coefficients of the branches whose U -order is d . More explicitly, a factor with root order below d contributes a nonzero constant to the weighted initial product, one above d contributes W , and one equal to d contributes $W - A$ with $A \neq 0$. The product has a nonzero root precisely in the last case.

Thus some branch is $V = \phi(U^{1/e})$ with $\text{ord}_Z \phi = de$. The construction in theorem 7.1 gives a nonzero positive-valuation solution at every chosen positive monomial scale. $\square$

Corollary 7.4 (Scale-independent existence). *For a fixed F over an algebraically closed characteristic-zero field, existence of a nonconstant solution is the same in every nonzero divisible Hahn value group. It is decided by finitely many tests (7.3): the finite diagonal-residue charts after removal of $Y - X$ factors, and the chart at infinity. For algebraic-number coefficients this is an effective decision procedure.*

Proof. The charts and their number are as in theorem 7.1. Each chart has a solution exactly when theorem 7.3 applies. The test uses polynomial arithmetic, roots of one ordinary polynomial, and comparison of finitely many integer weights. With exact algebraic coefficients, all the required equality tests and root calculations are effective. A nonzero divisible ordered group always has a positive element. $\square$

The criterion decides *existence*, not whether an arbitrary encoded infinite Hahn series satisfies the equation. It also does not make an arbitrary complex coefficient into computable input.

8 The exact classification in the negative-support ring

Theorem 8.1 (Polynomial-in-one-monomial criterion). *Let $\text{char } k = 0$, $d \geq 2$, and $y \in A_\Gamma(k) \setminus k$. The following are equivalent:*

- (i) y and S_dy are algebraically dependent over k ;
- (ii) $y = P(t^{-\delta})$ for some $\delta \in \Gamma_{>0}$ and $P \in k[T]$;
- (iii) the support of y is finite and its nonzero exponents generate a cyclic subgroup of Γ .

If a relation $F(y, S_dy) = 0$ is given, the polynomial in (ii) can be chosen of degree at most $\deg_Y F$.

Proof. Since $y \notin k$ and its support has no positive exponent, $v(y) < 0$. Put $\gamma = -v(y)$. By theorem 6.1, for some $e \leq \deg_Y F$ its support lies in $(\gamma/e)\mathbb{Z}$. It also lies between $-\gamma$ and zero. There are only $e + 1$ points of this cyclic subgroup in that interval. Consequently

$$y = \sum_{j=0}^e a_j t^{-j\gamma/e},$$

which proves (i) $\Rightarrow$ (ii) with the degree bound.

For (ii) $\Rightarrow$ (i), write $T = t^{-\delta}$. Then $y = P(T)$ and $S_dy = P(T^d)$ belong to the one-variable rational function field $k(T)$, whose transcendence degree is one. More explicitly, the nonzero polynomial

$$R_P(X, Y) = \text{Res}_T(P(T) - X, P(T^d) - Y) \quad (8.1)$$

vanishes on (y, S_dy) . Its degree in Y is $\deg P$, so it is not identically zero. Conditions (ii) and (iii) are equivalent by taking a positive generator of the subgroup generated by the finitely many support exponents. $\square$

Corollary 8.2 (Dilation independence). *For $y \in A_\Gamma(k)$, dependence of y, S_dy for one integer $d \geq 2$ is equivalent to dependence for every such integer. If y has infinite support, every pair y, S_dy is algebraically independent over k . The same independence conclusion holds for a finite support of rational rank at least two.*

Proof. The support condition in theorem 8.1 does not mention d . For a finite subset of a torsion-free abelian group, the generated subgroup is cyclic exactly when its rational span has dimension at most one. $\square$

Remark 8.3 (Why the support restriction is indispensable). A Laurent series can have infinitely many positive exponents even when it is cyclically supported. Intersecting with $A_\Gamma(k)$ cuts off all of them. Thus the theorem is not a cardinality argument. Its force comes from a *discreteness conclusion* followed by a *two-sided support bound*. A merely bounded, nondiscrete support can still be infinite.

9 Sharp algebraic degrees and recovery of a monomial

Definition 9.1. For a nonconstant finite-support $y \in A_\Gamma(k)$ satisfying the criterion, let $\delta > 0$ be the positive generator of the group generated by its nonzero exponents. Write $y = P(t^{-\delta})$. The polynomial P is *primitive in its exponents*: the greatest common divisor of its positive supported degrees is one. Define $M(y) = \deg P$.

Theorem 9.2 (The exact relation degree). *For every $d \geq 2$,*

$$\min_{\substack{0 \neq F \in k[X, Y] \\ F(y, S_dy) = 0}} \deg_Y F = M(y). \quad (9.1)$$

Moreover, if $T = t^{-\delta}$ is the primitive monomial, then

$$k(y, S_d y) = k(T). \quad (9.2)$$

The irreducible relation has bidegree $(\deg_X, \deg_Y) = (dM(y), M(y))$.

Proof. Let $M = M(y)$ and let a relation of Y -degree D be given. The degree-bounded conclusion of theorem 8.1 supplies $y = Q(t^{-\delta'})$ of degree $e \leq D$. The support-generated subgroup $\delta\mathbb{Z}$ is contained in $\delta'\mathbb{Z}$, so $\delta = m\delta'$ for an integer $m \geq 1$. Comparison of the largest support magnitude gives $M\delta = e\delta'$, hence $e = Mm \geq M$. Therefore $D \geq M$. The resultant (8.1) gives the reverse inequality.

For the field statement, the extension $k(T)/k(P(T))$ has degree M . One elementary proof is that $P(T) - X$ is irreducible in $k(X)[T]$: its ideal in $k[T, X]$ has quotient $k[T]$, and Gauss's lemma applies. If $L = k(P(T), P(T^d))$, the just-proved minimum degree says

$$[L : k(P(T))] = M.$$

The tower law then forces $L = k(T)$. Finally $[k(T) : k(P(T^d))] = dM$, which is the degree of the irreducible relation in X ; its degree in Y is M . $\square$

Corollary 9.3 (An ordinary polynomial field identity). *Let $P \in k[T] \setminus k$, and let $g = \gcd\{j \geq 1 : [T^j]P \neq 0\}$. In characteristic zero,*

$$k(P(T), P(T^d)) = k(T^g) \quad (d \geq 2). \quad (9.3)$$

Proof. Write $P(T) = Q(T^g)$, with Q primitive in its exponents, and apply theorem 9.2 to Q in an ordinary Laurent Hahn field. $\square$

This corollary can be read entirely in classical rational-function algebra. We give it as a consequence and consistency check of the support argument, not as a certified priority claim in polynomial decomposition theory. The restriction to polynomials is real: for $P(T) = T + T^{-1}$ one has $P(T^2) = P(T)^2 - 2$, and the generated field is $k(T + T^{-1})$, a proper subfield of $k(T)$.

Example 9.4 (A genuinely quadratic relation). For $y = T^2 + T$ and $d = 2$, the primitive degree is two. The relation

$$\begin{aligned} 0 &= Y^2 - (2X^2 + 6X + 2)Y \\ &\quad + X^4 + 2X^3 + 2X^2 \end{aligned} \quad (9.4)$$

is obtained by elimination and has the least possible Y -degree. There is no rational function $R \in k(X)$ with $S_2 y = R(y)$. Thus the implicit algebraic classification is strictly broader than the explicit rational-map classification in section 11.

10 Consequences for surreal and Gaussian omnific integers

Theorem 10.1 (Omnific dilation classification). *Let $d \geq 2$ and $x \in \mathbf{Oz} \setminus \mathbb{Z}$. The following are equivalent:*

- (i) $x, S_d x$ are algebraically dependent over $\mathbb{R}$;
- (ii) $x, S_d x$ are algebraically dependent over $\mathbb{C}$ inside $\mathbf{No}[i]$;
- (iii) for some $\delta \in \mathbf{No}_{>0}$, $x = P(\omega^\delta)$ with $P \in \mathbb{R}[T]$ and $P(0) \in \mathbb{Z}$;

(iv) x has finite normal-form support, and its positive exponents are commensurable.

With the primitive exponent step, its least relation degree is $\deg P$. For $x \in \mathbf{Oz}[i] \setminus \mathbb{Z}[i]$, the identical classification holds with $P \in \mathbb{C}[T]$, $P(0) \in \mathbb{Z}[i]$, and dependence over $\mathbb{C}$.

Proof. Choose a set-sized divisible subgroup of $\mathbf{No}$ containing the exponent support, and use (2.4). Apply theorems 8.1 and 9.2. In the real case, a complex polynomial relation splits into its real and imaginary coefficient polynomials; at least one is a nonzero real relation. Conversely a real relation is a complex one. The coefficient at zero imposes exactly the integer or Gaussian-integer condition and no condition on the other coefficients. $\square$

Corollary 10.2 (All distinct forward dilates). *If x is an infinite-support omnific or Gaussian omnific integer, then for every $d \geq 2$ and every pair of integers $0 \leq a < b$,*

$$S_d^a x \quad \text{and} \quad S_d^b x$$

are algebraically independent over $\mathbb{C}$ (and over $\mathbb{R}$ in the real case).

Proof. The first member still has infinite support and the second is its $S_{d^{b-a}}$ dilate. Apply theorem 10.1. $\square$

Example 10.3 (Finite but incommensurable). The number

$$x = \omega^{\sqrt{2}} + \omega + 1$$

is an omnific integer, and $x, S_d x$ are algebraically independent for every integer $d \geq 2$: the positive exponents $1, \sqrt{2}$ are $\mathbb{Q}$ -linearly independent. This particular x is also prime by the published theorem of L’Innocente–Mantova [14, Theorem B]. Its primality is cited, not proved or claimed as a contribution here. The dilation independence is a different property.

Example 10.4 (Comparable scales do not suffice). Both $\omega^{\sqrt{2}} + \omega$ and $\omega^{3/2} + \omega$ have two finite real exponents in the same Archimedean class of the exponent group. Only the second is a polynomial in one monomial: with $T = \omega^{1/2}$ it is $T^3 + T^2$. Its least relation degree is three; the first has no relation at all.

Example 10.5 (One rational span, unbounded denominators). The valid normal form

$$u_d = \sum_{n \geq 0} \omega^{d^{-n}}$$

has infinite support of order type ω . All its exponents lie in $\mathbb{Q}$, but not in any cyclic subgroup. It is algebraically independent of $S_d u_d$. In section 13 we show that its first three transforms satisfy an exact polynomial relation. Pairwise independence must not be confused with independence of a longer tuple.

Example 10.6 (A coefficient-field obstruction). Let $a \in \mathbb{R}$ be transcendental over $\mathbb{Q}$ and $\delta > 0$. Although $x = a\omega^\delta$ satisfies $S_d x = a^{1-d} x^d$ over $\mathbb{R}$, it has no first-order autonomous polynomial relation over $\mathbb{Q}$. This follows from corollary 7.2. The constants allowed in a relation materially change the answer.

11 Complete solutions for rational maps

We now classify the explicit equations $S_q y = R(y)$, with $q \in \mathbb{Q}_{>1}$, Γ divisible, and $R \in k(Z)$ nonconstant. The polynomial relation theorem allows implicit ramification; in this explicit case there is no such support ramification.

Theorem 11.1 (Rational-map classification). *Assume k algebraically closed of characteristic zero and $\Gamma \neq 0$ divisible. A nonconstant solution of*

$$S_q y = R(y) \quad (11.1)$$

exists exactly when R has a fixed point $c \in \mathbb{P}^1(k)$ whose local degree is q . In particular, q must be an integer $d \geq 2$. For every such fixed point, let

$$M_c(Y) = Y - c \quad (c \in k), \quad M_\infty(Y) = 1/Y,$$

and write

$$h = M_c \circ R \circ M_c^{-1} = AZ^d + O(Z^{d+1}).$$

All nonconstant solutions associated with c are exactly

$$y = M_c^{-1}(aH_a(t^\gamma)), \quad \gamma \in \Gamma_{>0}, \quad Aa^{d-1} = 1, \quad (11.2)$$

where H_a is the normalized inverse Böttcher coordinate of $a^{-1}h(aZ)$. Constant solutions are simply the finite fixed points of R where it is defined.

Proof. For $v(y) \geq 0$, let c be its residue. A pole of R at c would give $v(R(y)) < 0$, impossible. Reducing the equation modulo the maximal ideal therefore gives $R(c) = c$. For $v(y) < 0$, the equation requires ∞ to be a fixed point. In either case $u = M_c(y)$ is nonzero of positive valuation, and

$$S_q u = h(u).$$

If $m \geq 1$ is the local degree, comparison of valuations gives $q v(u) = m v(u)$, hence $q = m$. Since $q > 1$, this is an integer $d \geq 2$. The germ now has the form in proposition 5.2, which gives all solutions and their converse. Because k is algebraically closed, the required leading coefficients a exist. $\square$

In particular, not every nonlinear rational map admits a nonconstant solution. The relevant property is a *superattracting fixed point of exactly the dilation degree*, not just a nonlinear term somewhere in the formula. Over a non-algebraically-closed field the same description applies whenever the fixed point and the required leading-coefficient root lie in that field. For real solutions, these are genuine real-root conditions, not automatic consequences of the complex theorem.

Theorem 11.2 (Rational-map rigidity in the negative-support ring). *Let $\text{char } k = 0$, $y \in A_\Gamma(k) \setminus k$, and $d \geq 2$. If $S_d y = R(y)$ for $R \in k(Z)$, then*

$$y = \alpha t^{-\gamma} + b, \quad R(Z) = \alpha^{1-d}(Z - b)^d + b, \quad (11.3)$$

with $\alpha \in k^\times$, $b \in k$, and $\gamma \in \Gamma_{>0}$. Conversely every such pair satisfies the equation. For omnific solutions $b \in \mathbb{Z}$, and for Gaussian omnific solutions $b \in \mathbb{Z}[i]$.

Proof. Writing $R = A/B$ in reduced form, the equation $B(y)S_d y - A(y) = 0$ has degree one in the dilated variable. The denominator is nonzero since y is transcendental over k . By the degree bound in theorem 8.1, y is affine in one monomial, giving the first formula. Substitute this into the equation, and use transcendence of the monomial to obtain the displayed rational-function identity. The converse is direct substitution. The constant-term restrictions are exactly the definitions of the two omnific rings. $\square$

Corollary 11.3 (Simultaneous rational equations). *Suppose the same nonconstant $y \in A_\Gamma(k)$ satisfies $S_p y = P(y)$ and $S_q y = Q(y)$ for integers $p, q \geq 2$ and rational functions P, Q . Then, with one common α, b, γ ,*

$$y = \alpha t^{-\gamma} + b, \quad P(Z) = \alpha^{1-p}(Z - b)^p + b, \quad Q(Z) = \alpha^{1-q}(Z - b)^q + b.$$

In particular, P and Q commute. Even without the negative-support restriction, simultaneous nonconstant solutions force $P \circ Q = Q \circ P$.

Proof. Apply theorem 11.2; the representation is unique from its leading monomial and constant term. For the last assertion, commute the dilations and use the fact that they fix the coefficients: $(P \circ Q)(y) = (Q \circ P)(y)$. Transcendence of y turns equality at y into equality of rational functions. $\square$

12 A sharp infinitesimal-tail obstruction

The rational-map obstruction has a quantitative form. It identifies the first infinitesimal term that prevents an otherwise legitimate surreal solution from being an omnific integer.

Theorem 12.1 (First forbidden tail). *Let $P \in \mathbb{C}[Z]$ have degree $d \geq 2$, with leading coefficients p_d, p_{d-1} . Put*

$$b = -\frac{p_{d-1}}{dp_d}, \quad p_d \alpha^{d-1} = 1, \quad Q(X) = \alpha^{-1}(P(\alpha X + b) - b).$$

Then Q is monic and centered. If $Q \neq X^d$, write

$$Q(X) = X^d + cX^{d-r} + \text{lower powers}, \quad 2 \leq r \leq d, \quad c \neq 0.$$

For each positive scale γ , the infinite solution with leading coefficient α has expansion, with $T = t^{-\gamma}$,

$$y = \alpha T + b - \frac{\alpha c}{d} T^{1-r} + O(T^{-r}). \quad (12.1)$$

Its first positive-valuation term has valuation exactly $(r-1)\gamma$. In particular it does not belong to the negative-support ring.

Proof. The normalization eliminates the coefficient of X^{d-1} and makes the leading coefficient one. The infinity case of theorem 11.1 gives a unique normalized Laurent solution

$$F(T) = T + \sum_{n \geq 0} b_n T^{-n}, \quad F(T^d) = Q(F(T)),$$

with $y = \alpha F(T) + b$. Comparison at degree $d-1$ gives $b_0 = 0$. Successive comparison at degrees $d-2, d-3, \dots, d-r+1$ gives $b_1 = \dots = b_{r-2} = 0$. At degree $d-r \geq 0$, the left side has no term, and the right-side coefficient is $db_{r-1} + c$. Therefore $b_{r-1} = -c/d \neq 0$. This proves the expansion and the exact valuation. $\square$

Here $O(T^{-r})$ denotes an ordinary formal Laurent tail with powers at most $-r$; it is not an analytic estimate at all surreal magnitudes. After $T = t^{-\gamma}$, this notation has the exact support meaning required by the theorem.

12.1 Examples and exact coefficients

For $P(Z) = Z^2 + 1$, the unique infinite branch at each positive scale begins

$$y = T - \frac{1}{2}T^{-1} - \frac{3}{8}T^{-3} - \frac{3}{16}T^{-5} - \frac{45}{128}T^{-7} + \dots \quad (12.2)$$

It is a perfectly well-defined real surreal solution of $S_2y = y^2 + 1$ after $T = \omega^\gamma$, but is not omnific. There are no nonconstant omnific solutions to this equation.

For $P(Z) = Z^2 - 2$ there is an exact finite Laurent identity:

$$y = T + T^{-1}, \quad S_2y = T^2 + T^{-2} = y^2 - 2.$$

Again the infinitesimal term prevents an omnific solution. This example also explains why the full Hahn-field conclusion is Laurent-series rigidity, not polynomial rigidity.

For $P(Z) = Z^3 + Z$ one obtains

$$y = T - \frac{1}{3}T^{-1} - \frac{8}{81}T^{-5} - \frac{8}{243}T^{-7} + \dots$$

By contrast, $P(Z) = (Z - 3)^2 + 3$ has the omnific solutions $y = 3 + \omega^\gamma$, since its centered normalization is exactly Z^2 . The finite exact checks supplied with the article verify the displayed coefficients and the coordinate identities through the stated truncation orders; theorem 12.1 proves the infinite statement.

13 A strict hierarchy of autonomous algebraic orders

Definition 13.1. For a fixed dilation S_d , define the *autonomous algebraic order* of y over k to be the least $r \geq 0$ for which

$$y, S_dy, \dots, S_d^r y$$

are algebraically dependent over k . If no such r exists, set the order to $+\infty$. Constants have order zero. This definition concerns algebraic dependence of a transform tuple; it is not the usual order of an arbitrary variable-coefficient linear equation.

Theorem 13.2 (An infinite-support element of exact order two). *For each integer $d \geq 2$, the omnific integer*

$$u_d = \sum_{n \geq 0} \omega^{d^{-n}}$$

has exact autonomous algebraic order two over $\mathbb{C}$ for S_d . It satisfies

$$S_d^2 u_d - S_d u_d = (S_d u_d - u_d)^d. \quad (13.1)$$

Every two distinct forward transforms are nevertheless algebraically independent over $\mathbb{C}$.

Proof. Its decreasing normal-form exponents make the sum legitimate. Dilation is strong, and direct comparison of supports gives

$$S_d u_d = u_d + \omega^d.$$

Applying S_d once more gives $S_d^2 u_d - S_d u_d = \omega^{d^2} = (\omega^d)^d$, proving (13.1). The polynomial $Z - Y - (Y - X)^d$ is nonzero, so the order is at most two. By theorem 10.1, the infinite support makes $u_d, S_d u_d$ algebraically independent. Thus the order is exactly two. The last assertion is corollary 10.2. $\square$

Decreasing-denominator generalized series and their linear Mahler identities are already present in the literature; compare [10, Remark 2.18] and [9]. The series pattern is therefore not offered as a new construction. The point here is the rigorous *minimal autonomous order* and the resulting sharp boundary of the first-order classification in the omnific setting. [write] In the collection, the linear identity $S_2 u_2 - u_2 = \omega^2$ is the Conway form, after one application of S_2 , of the second series in the single-dilation report's example `dsup:ex:opposite` (section 1.5).

Theorem 13.3 (Every finite order occurs). *Let $g_1, \dots, g_r$ be positive surreal exponents linearly independent over $\mathbb{Q}$, and let $a_1, \dots, a_r \in \mathbb{R}^\times$. Then*

$$x = \sum_{j=1}^r a_j \omega^{g_j}$$

has exact autonomous algebraic order r over $\mathbb{C}$ for every integer $d \geq 2$.

Proof. Put $T_j = \omega^{g_j}$. The T_j are algebraically independent over $\mathbb{C}$, since distinct monomials in them have distinct normal-form exponents. For $0 \leq i < r$, put

$$Y_i = \sum_{j=1}^r a_j T_j^{d^i}.$$

Their Jacobian determinant is

$$\left(\prod_{j=1}^r a_j \right) \left(\prod_{i=0}^{r-1} d^i \right) \det(T_j^{d^i-1})_{0 \leq i < r, 1 \leq j \leq r}. \quad (13.2)$$

The last determinant is nonzero: its permutation terms have distinct exponent vectors, since the integers $d^i - 1$ are distinct. In characteristic zero, a nonzero Jacobian implies algebraic independence. For completeness, a polynomial relation of least total degree would, on differentiation, yield a nontrivial linear relation between the differentials dY_i . At least one evaluated partial derivative is nonzero, because otherwise a nonzero partial derivative would be a relation of smaller degree. This contradicts the determinant.

Thus the first r transforms are independent. The first $r + 1$ lie in the field $\mathbb{C}(T_1, \dots, T_r)$ of transcendence degree r , so they are dependent. The least order is exactly r . $\square$

Proposition 13.4 (Bounds from the rational rank of finite support). *Let x be a nonconstant finite-support omnific integer and let r be the dimension over $\mathbb{Q}$ of the span of its positive exponents. Its order is at most r . It is exactly one for $r = 1$ and exactly two for $r = 2$; for $r \geq 3$, the present results give the bounds $2 \leq \text{order} \leq r$.*

Proof. Choose a rational basis of the exponent span and clear the finitely many denominators. Then every transform of x lies in a rational function field generated by r algebraically independent monomials. This gives the upper bound. The first-order classification gives precisely the remaining assertions. $\square$

14 Boundaries that the conclusions cannot cross

14.1 Positive characteristic

If $k = \mathbb{F}_p$ and the dilation is S_p , then

$$S_p y = y^p$$

for every Hahn series y : Frobenius acts coefficientwise on these strong sums. Hence completely arbitrary well-ordered supports can satisfy a degree-one autonomous relation in the dilated variable. The characteristic-zero hypothesis cannot be dropped. In the proof, both Newton–Puiseux and the nonvanishing of the coefficient d are substantive.

14.2 A named nonconstant coefficient

The infinite-support series u_d satisfies the linear equation

$$S_d u_d - u_d = \omega^d.$$

The right side is not in the ordinary constant field. Thus there is no extension of the support-collapse theorem to unrestricted nonconstant coefficients. The distinction from variable-coefficient Mahler algorithms [11, 10] is structural, not a matter of proof technique. [write] At $d = 2$ this equation is the Conway form of `dsup:ex:opposite` after one application of S_2 (section 1.5).

14.3 Identity dilation and arbitrary rational dilations

For $d = 1$, every series satisfies $S_1 y - y = 0$, so the theorem is false. For a nonintegral rational $q > 1$, theorem 11.1 rules out explicit rational-map solutions. That argument does not classify implicit relations $F(y, S_q y) = 0$. In the Puiseux step the leading order becomes qe , and taking an e th root need not produce an ordinary integer-order germ. The proof given here must not be quoted as covering that case. [write] Implicit relations of this kind do have nonconstant solutions: by the single-dilation report’s `dsup:lem:rationalrecognition`, for divisible Γ and $q = a/b$ the nonzero solutions of $(S_q x)^b = x^a$ are exactly the monomials $\mu_{|a-b|}(k) t^\Gamma$. They satisfy the cyclic-support conclusion trivially; the general implicit case remains open (section 1.5).

14.4 Full first-order theories

The decision procedure below is for one specified autonomous equation with algebraic-number coefficients. It says nothing about arbitrary quantifier alternations in an expanded surreal field, arbitrary systems of omnific Diophantine equations, or equality of arbitrary infinite normal-form descriptions. In particular, it does not contradict the undecidability phenomena discussed in the repository’s single-dilation report [2].

15 Effective finite shapes and a decision procedure

The sharp degree bound turns the omnific problem into a finite algebraic problem, even though a general omnific integer is specified by a potentially transfinite normal form.

Theorem 15.1 (Finitely many bounded polynomial shapes). *Let k be algebraically closed of characteristic zero, $0 \neq F \in k[X, Y]$, and $d \geq 2$. Put $D = \deg_Y F$. The set*

$$\mathcal{P}_{\leq D}(F, d) = \{P \in k[T] : 1 \leq \deg P \leq D, F(P(T), P(T^d)) = 0\} \quad (15.1)$$

is finite. Every nonconstant solution in a negative-support Hahn ring has the form $P(t^{-\delta})$ with P in this finite set and $\delta > 0$. The analogous statement holds elementwise for omnific and Gaussian omnific integers, retaining the appropriate coefficient and constant-term conditions.

Proof. Fix a degree M . For a polynomial P of that degree satisfying the identity, view $P(t^{-1})$ in $k((t^{\mathbb{Q}}))$. It has valuation $-M$. By theorem 7.1 it equals $f(t^{\delta})$ for one of finitely many infinity profiles f of pole order e . Valuation comparison forces $\delta = M/e$. Thus, for each fixed M and profile, there is at most one possible polynomial P . Finiteness follows on taking $1 \leq M \leq D$. Exhaustion of all negative-support solutions is precisely the degree-bounded assertion of theorem 8.1 and transcendence of the chosen monomial. Filtering the coefficients gives the omnific versions. $\square$

The bound on the polynomial degree is essential to this finiteness statement. For example, $S_d y = y^d$ is satisfied by $P(T) = T^n$ for arbitrarily large n . These do not create new shapes modulo the free monomial scale, since $(t^{-\delta})^n = t^{-n\delta}$.

Theorem 15.2 (Decidable nonconstant omnific solvability). *Let $F \in \overline{\mathbb{Q}}[X, Y] \setminus \{0\}$ be supplied by exact algebraic-number coefficients and let $d \geq 2$. There is an algorithm to decide whether $F(x, S_d x) = 0$ has a nonconstant solution in $\mathbf{Oz}$, or in $\mathbf{Oz}[i]$. It produces a finite list of polynomial shapes whose positive surreal monomial evaluations give all such solutions. Every coefficient of those shapes is an algebraic number.*

Proof. If $D = \deg_Y F = 0$, no nonconstant solution exists. Otherwise, for each $1 \leq M \leq D$, introduce unknowns $a_0, \dots, a_M$ and form

$$P(T) = a_0 + a_1 T + \dots + a_M T^M.$$

Set every coefficient in T of $F(P(T), P(T^d))$ equal to zero and impose $a_M \neq 0$, for instance by adding an auxiliary inverse variable. These are finite polynomial systems with algebraic-number coefficients. By theorem 15.1, each has only finitely many solutions in $\mathbb{C}$. A finite algebraic set over $\overline{\mathbb{Q}}$ has algebraic-number coordinates. Its points can be computed by standard zero-dimensional polynomial-system methods. Alternatively, separate real and imaginary parts and use exact real-algebraic point isolation [15]; a nonzero complex leading coefficient is imposed by inverting its squared modulus.

For $\mathbf{Oz}$, retain precisely the shapes with every coefficient real and $a_0 \in \mathbb{Z}$. For $\mathbf{Oz}[i]$, retain those with $a_0 \in \mathbb{Z}[i]$. Reality, equality, and integrality of exact algebraic numbers are decidable. The retained list is nonempty exactly when a nonconstant solution exists. Its evaluations at $T = \omega^{\delta}$ give all solutions by theorem 10.1. A witness may always be produced already at $T = \omega$. $\square$

The algorithm is a termination result, not a practical complexity claim. The coefficient systems can be large, and a production implementation would exploit Puiseux branches and the finite-profile structure before using general elimination. The supplied Python program implements exact checks of representative identities, not this complete elimination algorithm.

15.1 A finite computational specification

The equation expanded in the proof has degree at most $M(\deg_X F + d \deg_Y F)$ in T . Thus there is an explicit finite bound on the number of coefficient equations. Every returned candidate is certified simply by the exact polynomial identity $F(P(T), P(T^d)) = 0$, the leading-coefficient test, and the constant-term condition. No truncation of an unknown infinite surreal number is used as a certificate of support termination.

The Newton-weight test of corollary 7.4 is a different algorithm: it decides nonconstant solvability in the full surcomplex field, where infinite Laurent tails are allowed. The two procedures can therefore legitimately give different answers, as for $S_2 x = x^2 + 1$.

16 Further research questions and priorities

The questions below are proposed research directions arising from this article. Unless explicitly stated otherwise, they are not claimed to be previously published named open problems. Their purpose is to specify concrete next theorems rather than to suggest that the present first-order classification covers a wider theory than it actually does.

Research question 1 (Exact order from finite support; high priority). For a finite-support omnific integer whose positive exponents have rational rank $r \geq 3$, is its autonomous order always r ? The answer is established here for $r \leq 2$ and for sums supported on a rationally independent set. The unresolved case allows several supported monomials in the same r -dimensional exponent span. A useful first target is rank three with four support terms. The bound in proposition 13.4 isolates the precise remaining gap.

Research question 2 (The support spectrum at order two; high priority). Which well-ordered Hahn supports can occur for omnific solutions of $F(x, S_d x, S_d^2 x) = 0$ over the ordinary constants? Infinite support of order type ω occurs by theorem 13.2. Determine whether fixed equation degree or fixed autonomous order bounds the possible support order types, denominator growth, or rational rank. The goal is a classification or a sharp no-bound theorem, not an extrapolation from the one telescoping example.

Research question 3 (Formal verification of the core theorem; high priority). Formalize the chain from strong Hahn evaluation through the monomial recognizer, the formal Böttcher coordinate, and the evaluated Puiseux branch to theorem 6.1. The most substantial imported obligation is a usable Newton–Puiseux theorem with the *individual branch* degree bound. A staged formalization can state that theorem as an explicit parameter before replacing it with a verified construction; it should not silently add it as an axiom and call the entire result verified.

Research question 4 (An efficient shape solver). Implement theorem 15.2 for rational or algebraic coefficients, using the Newton-weight filter, Puiseux branches, and polynomial degree bounds before general elimination. Prove bounds in terms of the input bidegree, coefficient height, and d . Can the finite candidate list be computed without constructing any nonpolynomial Böttcher profile? Can equivalent shapes under monomial rescaling be canonically minimized?

Research question 5 (Nonintegral rational dilation). Classify autonomous implicit equations $F(y, S_q y) = 0$ for $q \in \mathbb{Q}_{>1} \setminus \mathbb{Z}$. The explicit rational case has no nonconstant solutions, but a ramified correspondence can have the required rational Newton slope. Determine whether an appropriate generalized Böttcher coordinate still forces a controlled support subgroup, or permits unbounded denominators already at order one.

[write] *Status in the collection: open, with one partial datum.* The single-dilation report’s `dsup:lem:rationalrecognition` solves the particular relation $(S_q x)^b = x^a$, $q = a/b$, exactly: for divisible Γ its nonzero solutions are $\mu_{|a-b|}(k) t^\Gamma$. So the class of implicit relations with nonintegral q is not empty of nonconstant solutions, and for this relation the solutions are single monomials. Nothing is known here about general F .

Research question 6 (Tame positive-characteristic variants). When $\text{char } k = p > 0$ and $p \nmid d$, which support-collapse statements survive under restrictions on Puiseux ramification? Separate failure caused by wild algebraic branches from failure of the coordinate construction. The Frobenius example with $d = p$ shows that no unrestricted analogue is possible, but does not settle the tame cases.

Research question 7 (Systems and higher-dimensional correspondences). For a tuple $\mathbf{y}$ satisfying $S_d \mathbf{y} = \mathbf{R}(\mathbf{y})$, what bounds the number of independent support scales? In one dimension, a superattracting germ is formally conjugate to a monomial. Higher-dimensional germs need not have such a coordinate. Identify hypotheses under which a finite-rank support module, rather than one cyclic group, is forced.

Research question 8 (Controlled nonconstant coefficients). Replace the ordinary constant field by a finitely generated monomial field $k(t^\Delta)$. A linear equation already admits unbounded denominators, so cyclic support is too strong. Find the correct substitute: perhaps a bound on the support module generated under multiplication and division by d , relative to Δ , together with an ordinal support bound. Distinguish finite-generation assertions from mere countability.

Research question 9 (Arithmetic and convergence of the profiles). For algebraic coefficients, all profile coefficients are algebraic by corollary 7.2. Determine effective denominator and height bounds, and compare their Archimedean and non-Archimedean analytic convergence. The underlying Böttcher arithmetic is already an established subject [6]; the new target is a uniform statement through the ramified correspondence and the finite-profile list.

Research question 10 (Simultaneous algebraic correspondences). Describe the intersection of the finite-profile solution classes for two multiplicatively independent dilations. The rational omnific case is completely rigid by corollary 11.3, but implicit correspondences and full Laurent tails allow more geometry. Determine which compatibility conditions replace commutation of two rational maps.

Research question 11 (Definability of bounded shape complexity). For each fixed bound M , describe the locus of omnific elements whose least first-order relation degree is at most M , in the language with one named dilation and in weaker languages. The repository already recovers normal-form data from a dilation [2]; the question here is the complexity and uniformity of the *bounded-degree* stratification. Do not confuse a countable union of definable loci with one first-order definable set.

Research question 12 (Interaction with surreal exponential and derivation). The dilation in this article is defined by the normal-form exponent group, not by the surreal exponential. Study how the finite-profile classes behave under the surreal exponential and under specified surreal exponential-compatible derivations. Even a negative closure theorem would clarify which parts of difference algebra and differential algebra can be combined without changing the constant field or losing support control.

The recommended order is to formalize the support-collapse theorem, implement the finite-shape solver, and then attack the rank-three and order-two problems. Each has a precise success criterion and can be pursued without a general theory of analysis on the entire proper class.

17 Proof audit and concluding assessment

17.1 Dependency ledger

Input or conclusion	Status and dependency
Hahn field arithmetic and normal forms	Classical framework; used in set-sized workspaces.
Strong support lemma	Classical Neumann lemma; supplies all infinite formal evaluations.
Newton–Puiseux factorization	Classical characteristic-zero theorem; individual denominator bound proved here by the roots-of-unity orbit.
Formal Böttcher coordinate	Classical theorem; explicit logarithmic construction and uniqueness proof included.
Monomial recognizer	Already present for S_2 in the repository; general- d least-term proof included.
Support collapse and finite profiles	Proposed contribution; deductions from the preceding inputs, with no rank-one hypothesis.
Omnific criterion and sharp degree	Proposed consequences; follow by intersecting cyclic support with a bounded interval.
Finite-shape decision procedure	Uses the sharp degree bound, finite profiles, and standard exact finite algebraic-system solving.
Higher-order examples	Exact written identities and algebraic independence arguments; the generalized series pattern itself is prior.

The most delicate proof points have been made explicit: the selected Puiseux denominator is not a common splitting denominator; the root of the algebraic branch must be chosen to equal the actual $S_d z$; formal evaluation must respect composition and dilation; and the nondivisible-group descent uses a subgroup intersection containing the leading valuation. None of these is delegated to numerical evidence.

17.2 What has and has not been checked

The accompanying exact checks cover six inverse Böttcher expansions through degree 32 in the normalized power-series variable, twenty resultant examples, ten nonzero Jacobian instances, forty-five finite telescoping identities, four Frobenius examples, and six Newton-weight examples. All passed in the execution recorded in `verification_results.json`. The calculations use exact rational or polynomial arithmetic. Their purpose is to detect sign, normalization, indexing, and small-example errors.

These checks do not certify the full support theorem, the correctness of every classical dependency, or the decision procedure’s implementation. No Lean build was performed and no new Lean theorem is supplied. The repository’s pre-existing formalization status is not inherited by this article. Independent mathematical review is still needed.

The source search covered the named primary literature and the specified repository reports and inventory. The Nishioka–Nishioka paper was located and its abstract-level scope checked; its full text was not available through the retrieval route used here, and no detailed theorem from it is invoked. This is a limitation on novelty assessment, not a missing premise in any proof. An absence of matching search results is not proof of priority. [write] The full text of [7] was read when the manuscript joined the collection; what it contains, and the priority caveat that follows, are in Appendix C.3 and section 18.

17.3 Conclusion

The main structural distinction is now exact. At autonomous order one, an algebraic relation with an integer exponent dilation collapses an arbitrary Hahn support to a cyclic Laurent scale. In the omnific ring, that

collapse forces a finite polynomial in one monomial, and the primitive polynomial degree is recoverable as an algebraic extension degree. For a fixed equation there are only finitely many shapes modulo scale, giving an effective solvability reduction for algebraic input.

At order two the finite-support conclusion already fails. This sharp change is a useful organizing principle for further surreal difference algebra. It supplies both positive classifications and explicit negative results without relying on a problematic global topology or treating a proper class as an ordinary analytic field.

18 Consolidated ledger of non-claims

[write] The source states its limitations where they arise, and each is kept in place. This ledger collects them with their locations; “README” and “audit” refer to the delivered README (not shipped; its content is in the collection README) and to SOURCE_AUDIT.md. Items S1–S24 are the source’s; items W1–W7 were added when the report joined the collection.

- S1. Not refereed. No named published conjecture is claimed solved. Novelty is proposed, not certified; priority is not certified by an exhaustive literature review, and an absence of search results is not proof of priority (title page; section 17; README; audit).
- S2. The classical inputs are credited, not claimed: Hahn arithmetic and normal forms, the Neumann support lemma, Newton–Puisseux factorization, formal Böttcher coordinates and their arithmetic, the repository’s $d = 2$ recognizer, resultants and exact algebraic-system solving. Mahler equations, Böttcher coordinates and non-Puisseux solutions are not new topics (sections 1, 3, 5 and 17; audit).
- S3. The repository comparison covers selected reports and the inventory at the pin. It is not a line-by-line review of the repository and not an exhaustive repository-wide absence certificate; no clone or repository build was made and no repository file was changed (section 1; README; audit).
- S4. The headline equivalence is about polynomial relations with S_d over the ordinary constants; it does not say that an infinite omnific integer is algebraic over $\mathbb{R}$ (section 1).
- S5. This is not a factorization theorem and settles none of the remaining factorization conjectures for omnific integers. The primality of $\omega^{\sqrt{2}} + \omega + 1$ is cited, not proved or claimed; dilation independence is a different property (section 1, section 10; audit).
- S6. The encyclopedia page was used for orientation only, never in place of a primary source (section 1; audit).
- S7. The late-2025 preprints [12, 13] are cited from their abstracts for context, not as proof inputs (section 1; bibliography; audit).
- S8. The source compared Nishioka–Nishioka [7] at abstract level only; no result of it is a premise, and its full comparison was named as a priority task before any publication-level novelty claim (section 17; README; audit). See W2.
- S9. Membership in $k((t^\delta))$ is necessary, not sufficient, for a relation; the support conclusion does not turn strong evaluation into valuation convergence (the remark after theorem 6.1).
- S10. Coefficient descent is for a fixed coefficient field of the equation (after corollary 7.2).
- S11. The Newton-weight test decides existence, not whether an arbitrary encoded series satisfies the equation, and does not make arbitrary complex coefficients computable input; it decides solvability in the full field, so it can differ from omnific solvability, as for $S_2x = x^2 + 1$ (after corollary 7.4; section 15).

- S12. The polynomial-field identity corollary 9.3 is a consequence and consistency check, not a priority claim in polynomial decomposition theory; it fails for Laurent polynomials such as $T + T^{-1}$ (after corollary 9.3; audit).
- S13. Pairwise independence of dilates is not independence of a longer tuple (the example of u_d in section 10).
- S14. Over a field that is not algebraically closed, and for real solutions, the fixed point and leading-coefficient root in theorem 11.1 are genuine conditions, not automatic consequences of the complex theorem (after theorem 11.1).
- S15. In theorem 12.1, $O(T^{-r})$ is a formal Laurent tail, not an analytic estimate at all surreal magnitudes; the finite checks verify displayed coefficients through stated truncation orders only (after theorem 12.1; section 12).
- S16. The decreasing-denominator series pattern of u_d and its linear Mahler identity are prior [10, Remark 2.18], [9]; only the minimal autonomous order is claimed (after theorem 13.2; section 17; audit).
- S17. Characteristic zero cannot be dropped: in characteristic p with $d = p$, Frobenius gives arbitrary supports (section 14).
- S18. There is no extension to unrestricted nonconstant coefficients, by $S_d u_d - u_d = \omega^d$ (section 14).
- S19. For $d = 1$ the theorem is false; implicit relations $F(y, S_q y) = 0$ with nonintegral rational q are not covered by the proof (section 14). See W3.
- S20. The decision procedure concerns one specified equation with algebraic-number coefficients. It says nothing about full first-order theories, arbitrary systems of omnific Diophantine equations or equality of infinite normal-form descriptions, and does not contradict the single-dilation report’s undecidability (section 14).
- S21. The decision procedure is a termination result, not a complexity claim; it is not implemented, and `code/verify.py` checks representative identities only. The degree bound is essential to the finiteness of shapes (section 15; README; audit).
- S22. The exact autonomous order for finite support of rational rank $r \geq 3$ is open; only $2 \leq \text{order} \leq r$ is proved (proposition 13.4; section 16).
- S23. The twelve research questions are proposed directions, not previously published open problems. A formalization should not add the Newton–Puisseux theorem as an axiom and call the result verified (section 16; audit).
- S24. No Lean formalization is supplied and no repository Lean build was performed; the collection’s formalization status is not inherited. The finite exact checks cover finite instances only; they do not certify the infinite theorems, the classical dependencies or the decision procedure. Independent mathematical review remains necessary. The script should not be run with Python optimization flags that disable assertions (title page; section 17; Appendix B; README; audit).
- W1. No `adr:` label has a Lean mapping in `FORMALIZATION.md` (section 1.5).
- W2. The full text of [7], read for this report, records Mahler’s 1983 criterion [8] for a convergent power-series solution of a constant-coefficient relation $P(f(z), f(z^d)) = 0$. For ordinary power series over a number field or $\mathbb{C}$, existence of nonconstant solutions of such relations is therefore prior to this report, and no priority is claimed for theorem 7.3 and corollary 7.4 or theorem 7.1 restricted to that rank-one setting. Mahler’s paper itself was not read; its content is taken from [7] (Appendix C.3).

- W3. The single-dilation report's power-equation lemma gives nonconstant solutions of $(S_q x)^b = x^a$ for nonintegral q . This is a partial datum for the question on nonintegral rational dilation, which stays open (section 1.5, section 16).
- W4. None of the three questions of the single-dilation report (`dsup:sec:questions`) is answered; its question on definability from other exponent automorphisms is untouched beyond integer dilations. Its non-claim on variable-coefficient Mahler systems stands (section 1.5).
- W5. The exponent dilations here are not the argument dilations $f(z) \mapsto f(qz)$ of the holonomic-rigidity report, and nothing of that report is used or extended (section 1.5).
- W6. Nothing here bears on which value-group automorphisms admit exponential lifts (`q:image` of the exponential-automorphism report); S_d has no exponential lift by that report's `thm:dilation`. None of the surcomplex-automorphisms report's questions is addressed (section 1.5).
- W7. The source's repository statements were checked at the placement commit and stand; they are kept as provenance of its pin (Appendix C.2).

A A worked elimination identity

Let $X = T^2 + T$ and $Y = T^4 + T^2$. From $T^2 = X - T$ one obtains

$$T^4 = (X - T)^2 = X^2 + X - (2X + 1)T,$$

and hence

$$Y = X^2 + 2X - 2(X + 1)T.$$

Therefore

$$T = \frac{X^2 + 2X - Y}{2(X + 1)}.$$

The denominator is nonzero in $k(T)$ in characteristic zero. Substitution into $T^2 + T - X = 0$ and clearing denominators gives

$$Y^2 - (2X^2 + 6X + 2)Y + X^4 + 2X^3 + 2X^2 = 0.$$

This simultaneously exhibits the relation and reconstructs the primitive monomial from the pair. The degree theorem proves minimality; the calculation alone would not rule out a hidden linear relation.

B Reproducing the finite checks

The file `verify.py` requires Python 3.9 or later and SymPy. It uses only exact arithmetic. Run

```
python -m pip install sympy
python verify.py
```

The second command writes `verification_results.json` beside the script and prints the same report. The recorded run used Python 3.13.5 and SymPy 1.14.0. The script is not an implementation of infinite Hahn series and does not pretend that truncation is an infinite proof. [write] In this directory the script is `code/verify.py` and the recorded run is `data/verification_results.json`; see Appendix C.4 before rerunning.

For the inverse Böttcher checks, set $L(z) = zF(1/z)$ for a monic centered polynomial $Q(X) = \sum q_j X^j$ of degree d . The checked identity is

$$L(z^d) = \sum_{j=0}^d q_j z^{d-j} L(z)^j.$$

The coefficient of degree m on the right contains $d[z^m]L$ plus already known coefficients. Since $d \neq 0$, this is an exact recursion. The program independently recomputes the two sides after constructing the coefficients and verifies equality through degree 32.

For the finite sums $u_{d,N} = \sum_{n=0}^N \omega^{d^n}$, the checked identity is deliberately the correct truncated one,

$$S_d u_{d,N} - u_{d,N} = \omega^d - \omega^{d^N},$$

not the infinite identity with its boundary term silently deleted. The infinite identity follows separately from coefficientwise comparison of the full well-ordered supports.

C Provenance of this report

C.1 One manuscript

This report is built from *one* manuscript: the archive `surreal_dilation_rigidity`, manuscript 08 of batch 30 of the collection, dated 23 September 2026 and pinned to repository commit `b895e8672990e8b5a56f97dd7246f9dd5f86c771`. It was placed in commit `21375f8` as a new report: no report of the collection names its question, and it continues the single-dilation report's subject without answering any of that report's questions (section 1.5). There was no second source; nothing was merged and nothing was selected away. Every result, proof, example, question and limitation of the manuscript is printed here.

The delivered verification program `code/verify.py`, its recorded run `data/verification_results.json`, the dependency pin `data/requirements.txt` and the author-side audit `SOURCE_AUDIT.md` are kept byte-identical. The manuscript's article (delivered as `dilation_rigidity.tex`) and `README` were rewritten; the delivered PDF and checksum list are not shipped, and `article.pdf` is a build of this text.

The editorial changes are these.

- (1) Every label carries the prefix `adr:`. The source's 83 labels are kept, unchanged after the prefix, and eight were added (`adr:sec:conventions`, `adr:sec:collection`, `adr:sec:nonclaims`, `adr:app:report`, `adr:app:onesource`, `adr:app:pinned`, `adr:app:literature`, `adr:app:files`).
- (2) The title page names the report as a single-source report of the collection, gives the batch and pin, and extends the status box. Sections 1.4 and 1.5, the ledger of section 18 and this appendix are new, and the bibliography gained the entry [8]. Sentences marked [write] were added at the end of the source's subsection on the repository, at the end of its literature paragraph on Mahler equations, after lemma 3.3, after theorem 13.2, in section 14 (two places), after the question on nonintegral rational dilation, in section 17 and in Appendix B. The bibliography entry [7] gained one sentence.

No mathematical statement was changed, and no symbol was renamed: the collisions are between reports, and section 1.4 fixes the meanings. Where the edit had to choose: the manuscript was placed as a new report rather than as an addition to the single-dilation report, because it answers no question or named gap of that report; it was placed among the surcomplex reports, because its main theorem is uniform in the characteristic-zero coefficient field, its omnific corollary treats **Oz** and **Oz**[*i*] together, and its nearest neighbour is there; the title is kept; and the directory name records the subject, first-order autonomous relations between an element and its exponent dilate.

C.2 Repository statements at the pin

The source compared itself with the collection at its pin. Checked against the placement commit 21375f8, its statements stand.

- The single-dilation report uses the recognizer $S_2w = w^2$ (dsup:lem:monomialrecognition), treats linear and multiplicative dilation equations (dsup:thm:dilationcohom), proves undecidability of the expanded field (dsup:thm:undecidable), and recovers the normal-form data from one dilation (dsup:thm:definability).
- No report of the collection supplies the classification of theorems 6.1, 8.1 and 10.1; superattracting germs and Böttcher coordinates occur in no other report.
- The three reports the source inspected (single dilation, dynamics, entire functions at arbitrary rank) are unchanged between the pin and the placement commit, as are the surcomplex-automorphisms and exponential-automorphism reports compared in section 1.5. The holonomic-rigidity report gained material after the pin; none of it concerns exponent dilations.

C.3 References checked for this report

The placement audit confirmed the references to Neumann [4], Mannaa–Coquand [5], Salerno–Silverman [6], Chyzak–Dreyfus–Dumas–Mezzarobba [10] and Faverjon–Roques [11]. The following were checked when the report was written.

- *L’Innocente–Mantova* [14]. In the introduction of arXiv version 5, Theorem B states that the omnific integer $\omega^{\sqrt{2}} + \omega + 1$ is prime in **Oz**; the abstract describes this as Gonshor’s conjecture. The attribution in section 10 is therefore correct. The arXiv page lists the journal DOI 10.1016/j.aim.2024.109513; the volume number was not checked.
- *Nishioka–Nishioka* [7]. The bibliographic data were confirmed on Project Euclid, and the full text (seven pages, open access) was read. It works with ordinary convergent power series over a number field or $\mathbb{C}$ and proves transcendence of values. It records Mahler’s 1983 criterion [8] (Theorem 2 there): a convergent power series $f(z) = f_0 + f_r z^r (1 + \sum_{j \geq 1} f_j z^j)$ satisfying $P(f(z), f(z^d)) = 0$, for an irreducible P with algebraic-number coefficients that is not a constant multiple of $u - v$, exists if and only if three conditions on a Puiseux expansion of the curve $P = 0$ at a point (f_0, f_0) hold. It also records S. Nishioka’s algebraic independence theorem for solutions of first-order rational Mahler equations of distinct degrees; the example $f(z^2) = f(z)^2 / (1 - 2f(z)^2)$ with solution $(z^r + z^{-r})^{-1}$, whose reciprocal is the identity $y = T + T^{-1}$, $S_2 y = y^2 - 2$, of section 12; and a computation showing that $g(z^d) = g(z)^d + c$ with $d \geq 3$ and $c \neq 0$ has no solution $g \in \mathbb{C}(z)$ (the proof first reduces to Laurent polynomials), a rank-one relative of theorems 11.2 and 12.1. It contains nothing on Hahn series of higher rank, support collapse, negative-support rings, relation degrees or omnific integers. Mahler’s criterion is a rank-one antecedent of the existence statements in theorem 7.3 and corollary 7.4 and of theorem 7.1, restricted to ordinary power series (non-claim W2). Mahler’s paper itself was not read.
- *Gontsov–Goryuchkina* [9]. Title, journal, volume, year and article number were confirmed through the DOI registry; the paper was not read, so the source’s use of it for the decreasing-denominator pattern is unverified here.
- *Faverjon–Poulet* [12] and *Faverjon–Roques* [13]. Titles, authors and arXiv numbers were confirmed on arXiv (submitted 24 November and 11 December 2025). Only the abstracts were read, as by the source.

The first abstract describes an algorithm computing a basis of solutions of any Mahler equation, with a decomposition over Puiseux series. The second describes a purity theorem, that certain arithmetic properties of a Mahler function propagate to all solutions of its minimal equation; the source’s summary of it in section 1 as developing generalized Mahler-series structure could not be confirmed from the abstract. Neither is a proof input.

- Basu–Pollack–Roy [15] and the encyclopedia page [16] were not checked.

C.4 Files as placed and rerun

File	Role
code/verify.py	the delivered finite checks (delivered as verify.py)
data/verification_results.json	the delivered recorded run
data/requirements.txt	the delivered pin sympy==1.14.0
SOURCE_AUDIT.md	the delivered author-side source and claim audit

The program writes `verification_results.json` beside itself, overwriting any file of that name, and prints the same report. In this layout a run in place would create `code/verification_results.json` inside the report directory. Appendix B and the audit describe the flat delivered layout. Run it on a copy, from the report directory, with a fresh scratch directory `$T`:

```
python -m pip install -r data/requirements.txt
cp code/verify.py $T/ && (cd $T && python verify.py)
```

and compare `$T/verification_results.json` with `data/verification_results.json`. This was done for this report under Python 3.14.4 with SymPy 1.14.0: exit code 0, status PASS, and the output agrees with the recorded run except for the field recording the Python version (3.13.5 in the recorded run).

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